

Reading sheets

INTRODUCTION TO IMPORTING DATA IN R



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XLConnect

- Martin Studer
- Work with Excel through R
- Bridge between Excel and R
- XLS and XLSX
- Easy-to-use functionality

Installation

```
install.packages("XLConnect")
```

```
also installing the dependencies 'XLConnectJars', 'rJava'  
...
```

- Problems?
 - Install Oracle's Java Development Kit (JDK)
 - Google your error!

loadWorkbook()

```
library("XLConnect")  
book <- loadWorkbook("cities.xlsx")  
str(book)
```

```
Formal class 'workbook' [package "XLConnect"] with 2 slots  
  ..@ filename: chr "cities.xlsx"  
  ..@ jobj      : ...
```

getSheets()

```
getSheets(book)
```

```
"year_1990" "year_2000"
```

```
library(readxl)  
excel_sheets("cities.xlsx")
```

```
"year_1990" "year_2000"
```

readWorksheet()

```
readWorksheet(book, sheet = "year_2000")
```

```
      Capital Population
1 New York    17800000
2   Berlin    3382169
3   Madrid    2938723
4 Stockholm    1942362
```


readWorksheet()

Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

year_2000 col 2

row 3
row 4

```
readWorksheet(book, sheet = "year_2000",  
             startRow = 3,  
             endRow = 4,  
             startCol = 2,  
             header = FALSE)
```

```
Col1  
1 3382169  
2 2938723
```


Let's practice!

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Adapting sheets

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New data!

```
pop_2010 <- data.frame(Capital = c("New York", "Berlin", "Madrid", "Stockholm"),  
  Population = c(8191900, 3460725, 3273000, 1372565))
```

```
pop_2010
```

	Capital	Population
1	New York	8191900
2	Berlin	3460725
3	Madrid	3273000
4	Stockholm	1372565

createSheet()

```
pop_2010 <- ... # truncated  
library(XLConnect)  
book <- loadWorkbook("cities.xlsx")
```

Capital	Population
New York	16044000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

year_1990

Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

year_2000



createSheet()

```
pop_2010 <- ... # truncated
library(XLConnect)
book <- loadWorkbook("cities.xlsx")
createSheet(book, name = "year_2010")
```

The diagram illustrates the concept of a 'wide' table. It shows a single table with 4 columns: Capital, Population, year_1990, and year_2000. This is contrasted with a 'long' table structure where the same data is spread across multiple rows, with columns for Capital, Population, and year (1990, 2000, 2010). An Excel icon is present in the top right corner.

Capital	Population	year_1990	year_2000
New York	16044000		
Berlin	3500000		
Madrid	3000000		
Stockholm	1000000		

Capital	Population	year
New York	17800000	1990
Berlin	3500000	1990
Madrid	2800000	1990
Stockholm	1000000	1990
New York	20000000	2000
Berlin	3500000	2000
Madrid	3000000	2000
Stockholm	1000000	2000
New York	20000000	2010
Berlin	3500000	2010
Madrid	3000000	2010
Stockholm	1000000	2010

writeWorksheet()

```
pop_2010 <- ... # truncated
library(XLConnect)
book <- loadWorkbook("cities.xlsx")
createSheet(book, name = "year_2010")
writeWorksheet(book, pop_2010, sheet = "year_2010")
```

The diagram shows a transformation of a single table into a wide table structure. On the left, a single table lists the population of four cities in 1990:

Capital	Population
New York	16044000
Berlin	3500000
Madrid	3000000
Stockholm	1000000

Below this table is the label "year_1990". An Excel icon is positioned to the right of the first table. In the center, a second table lists the same four cities in 2000:

Capital	Population
New York	17800000
Berlin	3500000
Madrid	2900000
Stockholm	1000000

Below this table is the label "year_2000". On the right, a wide table structure is shown, representing the data for the year 2010. It has two columns and four rows, corresponding to the cities listed in the previous tables. Below this wide table is the label "year_2010".

saveWorkbook()

```
pop_2010 <- ... # truncated
library(XLConnect)
book <- loadWorkbook("cities.xlsx")
createSheet(book, name = "year_2010")
writeWorksheet(book, pop_2010, sheet = "year_2010")
```



Capital	Population
New York	16044000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

year_1990

Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

year_2000

Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

year_2010

saveWorkbook()

```
pop_2010 <- ... # truncated
library(XLConnect)
book <- loadWorkbook("cities.xlsx")
createSheet(book, name = "year_2010")
writeWorksheet(book, pop_2010, sheet = "year_2010")

saveWorkbook(book, file = "cities2.xlsx")
```

Capital	Population
New York	16044000
Berlin	3382169
Madrid	2938723
Stockholm	1942362
year_1990	
Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362
year_2000	
Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362
year_2010	

 cities2.xlsx

renameSheet()

```
renameSheet(book, "year_1990", "Y1990")
renameSheet(book, "year_2000", "Y2000")
renameSheet(book, "year_2010", "Y2010")
```

Capital	Population
New York	16044000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

year_1990

Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

year_2000

Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

year_2010



renameSheet()

```
renameSheet(book, "year_1990", "Y1990")
renameSheet(book, "year_2000", "Y2000")
renameSheet(book, "year_2010", "Y2010")
saveWorkbook(book, file = "cities3.xlsx")
```

Capital	Population
New York	16044000
Berlin	3382169
Madrid	2938723
Stockholm	1942362
Y1990	
Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362
Y2000	
Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362
Y2010	

 cities3.xlsx

removeSheet()

```
removeSheet(book, sheet = "Y2010")
```

Capital	Population
New York	16044000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

Y1990

Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

Y2000

Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362

Y2010



removeSheet()

```
removeSheet(book, sheet = "Y2010")  
saveWorkbook(book, file = "cities4.xlsx")
```

Capital	Population
New York	16044000
Berlin	3382169
Madrid	2938723
Stockholm	1942362
Y1990	
Capital	Population
New York	17800000
Berlin	3382169
Madrid	2938723
Stockholm	1942362
Y2000	



Wrap-up

- Basic operations
- Reproducibility is the key!
- More functionality
 - Styling cells
 - Working with formulas
 - Arranging cells
 - ...

Let's practice!

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